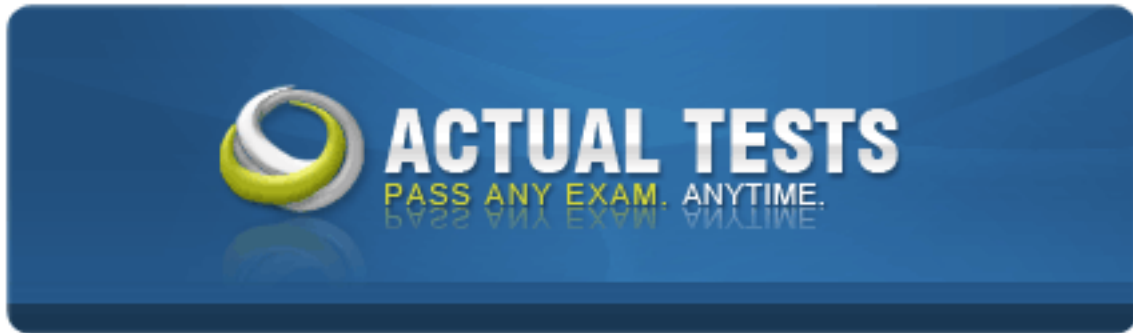


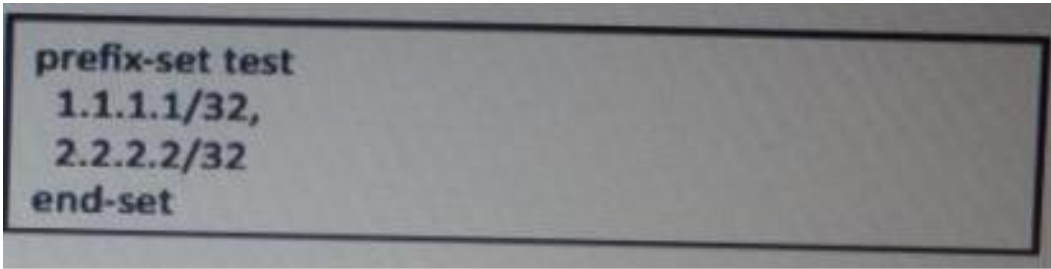
Cisco 642-775



Maintaining Cisco Service Provider Routing Protocols (MSPRP)

Version: 4.1

QUESTION NO: 1



```
prefix-set test
1.1.1.1/32,
2.2.2.2/32
end-set
```

Refer to the exhibit. Which commands append a prefix of 3.3.3.3/32 to prefix-set "test"?

- A.** Prefix-set test
3.3.3.3/32
End-test
- B.** Prefix-set test
Append 3.3.3.3/32
End-set
- C.** Prefix-set test
1.1.1.1/32,2.2.2.2/32,3.3.3.3/32
End-set

Answer: C

Explanation:

QUESTION NO: 2

Which statement accurately describes the BGP community filtering in Cisco IOS Software?

- A.** can be configured for IBGP neighbors only
- B.** can be configured for both IBGP and EBGP neighbors
- C.** can be configured for EBGP neighbors only
- D.** cannot be configured for IBGP or EBGP neighbors

Answer: B

Explanation:

QUESTION NO: 3

What effect will the command `timers throttle spf 5 1000 90000` have on an OSPF router?

- A.** The router dynamically reduces the frequency with which it computes the SPF tree during times of network instability.
- B.** The command improves router performance by limiting the amount of CPU cycles that are consumed over time when calculating the SPF tree.
- C.** When recalculating the SPF tree, the LSA update timers are changed to lessen the impact on memory and CPU usage during LSA floods.
- D.** The command specifies the number of LSAs that must be received in a particular time frame before initiating a complete SPF recalculation.

Answer: A

Explanation:

QUESTION NO: 4

While troubleshooting a loss of connectivity, you notice the log entry

Mar 16 08:20:16 %OSPF-5-ADJCHG: Process 1. Nbr 1.1.1.130 on Vlan30 from FULL to EXSTART. SeqNumberMismatch

After logging into the device, you notice that the router seems to be stuck in the EXSTART state

When does this problem most frequently occur?

- A.** during startup of the OSPF process
- B.** during high CPU load on the router
- C.** when the two routers report different versions of OSPF
- D.** when there is a mismatch between MTU settings for neighbor router interfaces

Answer: D

Explanation:

QUESTION NO: 5

Which Cisco IOS XR Software hierarchical Route Policy Language policy configuration uses the existing RPL policy "test"?

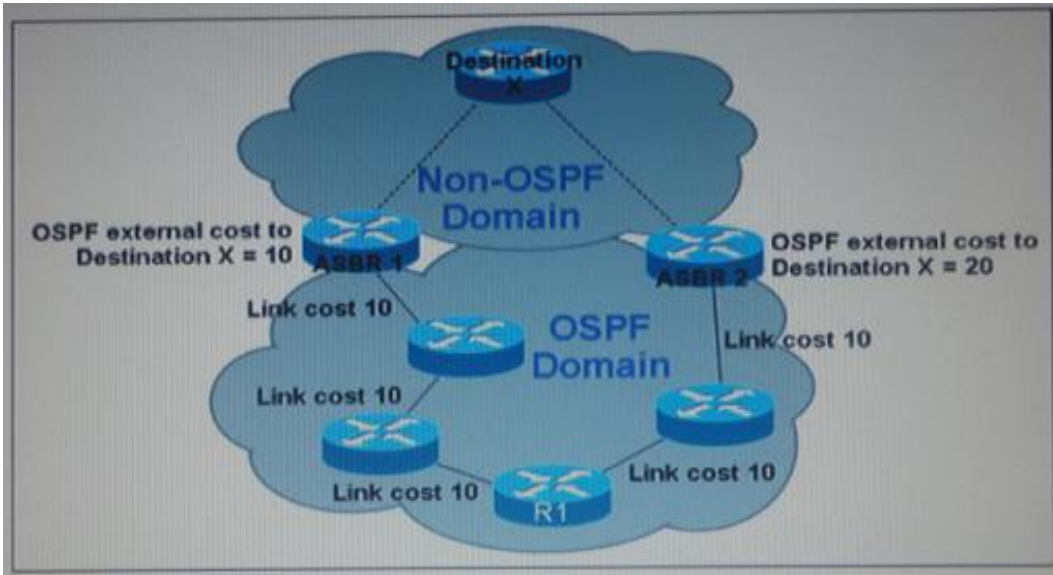
- A.** route-policy test
apply test
- B.** route-policy new
use test
- C.** route-policy new
apply test

D. route-policy new
use policy test

Answer: C

Explanation:

QUESTION NO: 6



Refer to the exhibit. What is the about destination x from the point of view of R1 if both ASBRs are advertising Destination x as an external type route?

- A. R1 sees each path as equal cost and load balances data traffic on both paths.
- B. R1 sees each path as equal cost and choose the path through the ASBR with the lower router ID
- C. R1 sees each path as ASBR1 as the shortest path
- D. R1 sees each path as ASBR2 as the shortest path

Answer: C

Explanation:

QUESTION NO: 7

How often are complete sequence number PDUs sent by an IS-IS router to its neighbor on a point to point link?

- A. Once when the link comes up, and every 10 seconds thereafter

- B. Once when the link comes up, and every 30 seconds thereafter.
- C. Once when the link comes up, and every 60 seconds thereafter.
- D. Once when the link comes up, and whenever there is a topology change

Answer: A

Explanation:

QUESTION NO: 8

An ISP is using OSPF as the IGP with a single OSPF area. The router memory usage for OSPF is too high. Which method can lower the OSPF memory usage?

- A. implement multiple OSPF areas
- B. implement BGP synchronization to synchronize between OSPF and BGP
- C. implement OSPF sham links
- D. implement passive interfaces

Answer: A

Explanation:

QUESTION NO: 9

Which keyword will stop further processing of the router policy language policy on a cisco IOS router?

- A. Stop
- B. Quit
- C. Drop
- D. End
- E. Exit

Answer: D

Explanation:

QUESTION NO: 10

What is the default OSPF hello interval when configured on a point-to-point interface in cisco IOS software?

- A. 3 seconds
- B. 10 seconds
- C. 15 seconds
- D. 30 seconds

Answer: B

Explanation:

QUESTION NO: 11

Which command gathers information on BGP route instability in Cisco IOS XR Software?

- A. show ip bgp summary
- B. show bgp flap-statistics
- C. show ip bgp neighbors
- D. show bgp ip dampening

Answer: B

Explanation:

QUESTION NO: 12

Which two statements correctly describe the IBGP split-horizon rule?

- A. Updates that are received on an EBGP session are forwarded to IBGP and EBGP.
- B. Updates that are received on an EBGP session are not back out the EBGP peers that are in the same neighboring AS.
- C. Updates that are received on an IBGP session are forwarded to all EGBP peers
- D. Updates that are received on an IBGP session can be forwarded to other IBGP peers synchronization is turned off.

Answer: A,B

Explanation:

QUESTION NO: 13

Which four BGP parameters can be tuned to influence BGP convergence time? (Choose four)

- A. scan time
- B. stuck in active timer
- C. advertisement interval
- D. keepalive interval
- E. holdtime interval
- F. idle time

Answer: A,C,D,E

Explanation:

QUESTION NO: 14

Which statement accurately describes the BGP community filtering in Cisco IOS XR Software?

- A. can be configured for IBGP neighbors only
- B. can be configured for both IBGP and EBGP neighbors
- C. can be configured for EBGP neighbors only
- D. cannot be configured for IBGP or EBGP neighbors

Answer: B

Explanation:

QUESTION NO: 15

If the network is experiencing high CPU usage that is caused by OSPF, which four options might you consider to reduce the CPU usage? (Choose four)

- A. Enable OSPF incremental Shortest Path First.
- B. Enable OSPF Shortest Path First throttling.
- C. Enable multiple OSPF areas.
- D. Enable auto-summary.
- E. Enable special OSPF area types (stub, totally stubby).
- F. Enable the "broadcast" OSPF network type.

Answer: A,B,C,E

Explanation:

QUESTION NO: 16

===== r1 =====

*Mar 10 15:24:25.807 UTC: OSPF: Send hello to 224.0.0.5 area 0 on Ethernet0/0 from 65.255.56.1

*Mar 10 15:24:27.171 UTC: OSPF: rcv. v:2 t:1 l:44 rid:65.255.255.4 ald:0.0.0.0 chk:30A1 aut:0 auk: from Ethernet0/0

*Mar 10 15:24:27.171 UTC: OSPF: Rcv hello from 65.255.255.4 area 0 from Ethernet0/0 65.255.56.2

*Mar 10 15:24:27.171 UTC: OSPF: Mismatched hello parameters from 65.255.56.2

*Mar 10 15:24:27.171 UTC: OSPF: Dead R 40 C 40, Hello R 10 C 10 Mask R 255.255.255.248 C 255.255.255.252

===== r2 =====

*Mar 10 15:24:46.251 UTC: OSPF: Send hello to 224.0.0.5 area 0 on Ethernet0/0 from 65.255.56.2

*Mar 10 15:24:54.471 UTC: OSPF: rcv. v:2 t:1 l:44 rid:65.255.255.3 ald:0.0.0.0 chk:309F aut:0 auk: from Ethernet0/0

*Mar 10 15:24:54.471 UTC: OSPF: Rcv hello from 65.255.255.3 area 0 from Ethernet0/0 65.255.56.1

*Mar 10 15:24:54.471 UTC: OSPF: Mismatched hello parameters from 65.255.56.1

*Mar 10 15:24:54.471 UTC: OSPF: Dead R 40 C 40, Hello R 10 C 10 Mask R 255.255.255.252 C 255.255.255.248

Refer to the exhibit. Two Cisco IOS routers that are configured with OSPF cannot establish adjacency. Examine the information that is collected from r1 and r2. Why cannot the adjacency be established?

- A. Network-type mismatch
- B. Different MTU
- C. Subnet mismatch
- D. Area ID mismatch

Answer: C

Explanation:

QUESTION NO: 17

```
RP/0/RP0/CPU0:crs-1#show int TenGigE0/4/4/0
TenGigE0/4/4/0 is up, line protocol is up
Interface state transitions: 2
Hardware is TenGigE, address is 001b.5375.5058 (bia 001b.5375.5058)
Internet address is Unknown
MTU 1500 bytes, BW 10000000 Kbit
  reliability 255/255, txload 11/255, rxload 23/255
Encapsulation ARPA,
Full-duplex, 10000Mb/s, LR, link type is force-up
output flow control is off, input flow control is off
loopback not set,
```

Refer to the exhibit. The NOC has informed you about a problem on a Gigabit Ethernet link between a newly installed CRS-1 that is running Cisco IOS XR Software and Cisco IOS router. These two routers cannot establish OSPF adjacency. When you connected to CRS-1, you noticed that the Gigabit Ethernet interface MTU is 1500. You must adjust the interface MTU on the Cisco IOS router.

Which MTU should you configure on the Cisco IOS router interface to match the MTU that is configured on the Cisco IOS XR software?

- A. 1500
- B. 1486
- C. 1486
- D. 1488

Answer: A

Explanation:

QUESTION NO: 18

Which list does the SNMP object "ospflfTable" provide?

- A. Internet-facing routes
- B. IGP fast-timers
- C. OSPF-to-EGP mappings
- D. interfaces from the viewpoint of OSPF

Answer: D

Explanation:**QUESTION NO: 19 DRAG DROP**

You want to make sure that all Telnet packets that are generated by the local router are marked with TOS 5. Configure the router by using the commands on the left so that all Telnet packets, regardless of the outgoing interface, are marked with TOS 5. Drag the command to the proper order on the right. Not all options are used.

set ip tos 5	first command
access-list 100 permit tcp any any eq telnet	second command
access-list 100 ip any any eq telnet	third command
route-map local-policy permit 10	fourth command
route-map local-policy deny 10	fifth command
match ip address 100	
match ip access-list 100	
ip local policy route-map local-policy	
ip policy route-map local-policy	

Answer:

You want to make sure that all Telnet packets that are generated by the local router are marked with TOS 5. Configure the router by using the commands on the left so that all Telnet packets, regardless of the outgoing interface, are marked with TOS 5. Drag the command to the proper order on the right. Not all options are used.

set ip tos 5	access-list 100 permit tcp any any eq telnet
access-list 100 permit tcp any any eq telnet	route-map local-policy permit 10
access-list 100 ip any any eq telnet	match ip address 100
route-map local-policy permit 10	set ip tos 5
route-map local-policy deny 10	ip policy route-map local-policy
match ip address 100	
match ip access-list 100	
ip local policy route-map local-policy	
ip policy route-map local-policy	

Explanation:

access-list 100 permit tcp any any eq telnet – first command

route-map local-policy permit 10 – second command

match ip address 100 – third command

set ip tos 5 – fourth command

ip local policy route-map local-policy – fifth command

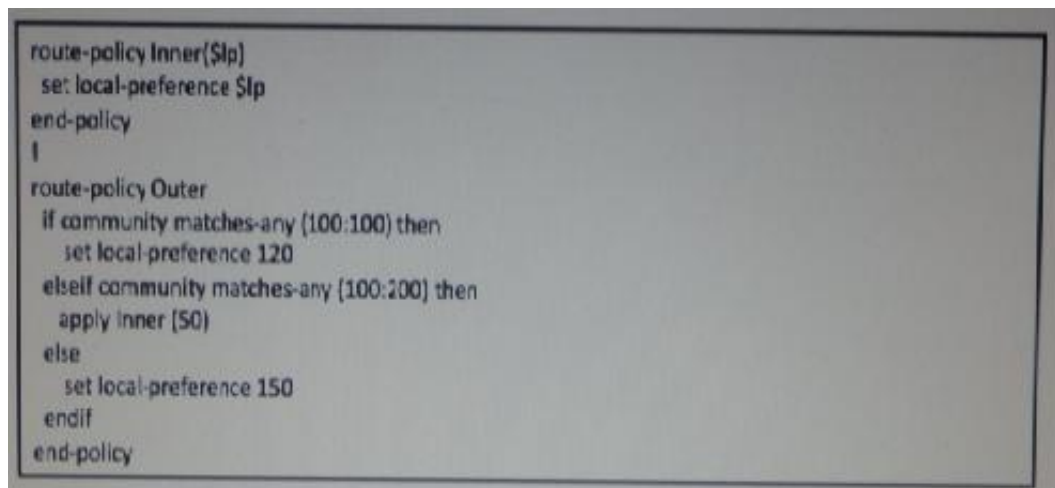
QUESTION NO: 20

For a CRS-1 router that runs Cisco IOS XR Software Release 3.6.6 in an OSPF core. Which protocol or feature improves router availability and can operate independently of remote router protocol support?

- A. BFD
- B. NSF IETF
- C. NSR
- D. PIM
- E. IGMP
- F. MPLS LDP sync
- G. MPLS-TE FRR

Answer: C

Explanation:

QUESTION NO: 21

```
route-policy Inner($lp)
  set local-preference $lp
end-policy
!
route-policy Outer
  if community matches-any (100:100) then
    set local-preference 120
  elseif community matches-any (100:200) then
    apply inner (50)
  else
    set local-preference 150
  endif
end-policy
```

Refer to the exhibit. What will the local-preference value for a route with a community value 100:200 if the Cisco IOS XR software route policy language policy “Outer” is applied?

- A. 50
- B. 120
- C. 150

- D. 200
- E. Will not change

Answer: A

Explanation:

QUESTION NO: 22

Refer to the exhibit.

```

PE1#show run | begin router ospf 1
router ospf 1
 log-adjacency-changes
 redistribute static subnets
 network 10.0.0.0 0.255.255.255 area 0
 network 192.168.0.0 0.0.255.255 area 0
!
<output omitted>

PE1#show interfaces GigabitEthernet0/0
GigabitEthernet0/0 is up, line protocol is up
Hardware is RV6460 Internal MAC, address is 0026.cbf5.a31b
(bia0026.cbf5.a31b)
Description: Link to P1
Internet address is 192.168.143.44/24
MTU 4500 bytes, BW 1000000 Kbit, DLY 10 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full Duplex, 100Mbps, media type is RJ45
output flow-control is XON, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:05, output 00:00:00, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
209810 packets input, 24838502 bytes, 0 no buffer
Received 191672 broadcasts (0 IP multicasts)
0 runs, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 watchdog, 234170 multicast, 0 pause input
245364 packets output, 23638651 bytes, 0 underruns
2 output errors, 0 collisions, 1 interface resets
0 babbles, 0 late collision, 0 deferred
2 lost carrier, 0 no carrier, 0 pause output

RP/0/0/CPU0:PE4#show run | begin router ospf 1
router ospf 1
 address-family ipv4 unicast
 area 0
 interface Loopback0
 passive enable
!
 interface GigabitEthernet0/4/0/2
 cost 1000
!
 interface GigabitEthernet0/4/0/3
 cost 1000
!
!
<output omitted>

RP/0/0/CPU0:PE4#show interfaces GigabitEthernet0/4/0/2
GigabitEthernet0/4/0/2 is up, line protocol is up
Interface state transitions: 16
Hardware is GigabitEthernet/IEEE 802.3 interface(s), address is
0018.18f7.4792
(bia 0018.18f7.4792)
Internet address is 192.168.143.45/24
MTU 4500 bytes, BW 1000000 Kbit
reliability 255/255, txload 0/255, rxload 0/255
Encapsulation ARPA,
Full-duplex, 1000Mb/s, SFPD, link type is force-up
output flow control is on, input flow control is off
loopback not set,
ARP type ARPA, ARP timeout 04:00:00
Last clearing of "show interface" counters never
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
0 packets input, 0 bytes, 0 total input drops
0 drops for unrecognized upper-level protocol
Received 0 broadcast packets, 0 multicast packets
0 runs, 0 giants, 0 throttles, 0 parity
60 packets output, 13560 bytes, 0 total output drops
Output 0 broadcast packets, 60 multicast packets
0 output errors, 0 underruns, 0 applique, 0 resets
0 output buffer failures, 0 output buffers swapped out
15 carrier transitions
  
```

The Cisco IOS software platform PE1 has an interface Gigabit Ethernet 0/0 that is connected to the Cisco IOS XR platform PE4 interface Gigabit Ethernet 0/4/0/2. OSPF cannot establish an adjacency. What is possible solution to the problem?

- A. Explicitly configure the cisco IOS XR software interface to be the active OSPF interface.
- B. Configure OSPF on each platform to ignore the interface MTU values.
- C. Adjust the Cisco IOS XR software hello and dead times to match the Cisco IOS software default timers
- D. Change the interface configurations on both platforms to point-to-point network type.

Answer: B

Explanation:

QUESTION NO: 23

A router that runs Cisco IOS Software Release 12 2(33) SXH generates a trap with a "cbgpBackwardTranransaction" message. What does this message indicate?

- A. BGP FSM state change has occurred and is causing transition errors.
- B. The BGP FSM moved from a higher-numbered state to a lower-numbered state.
- C. The BGP peer remote address was dropped during transition
- D. A BECN notification has caused a BGP peer transition.

Answer: B

Explanation:

QUESTION NO: 24 DRAG DROP

Drag the command on the left to the proper order on the right to build a route map to redistribute OSPF prefixes into BGP, so that certain prefixes that are tagged with 65001 are not redistributed.

route-map ospf-to-bgp deny 10	first command
route-map ospf-to-bgp permit 10	second command
match ip tag 65001	third command
match tag 65001	
route-map ospf-to-bgp permit 20	
route-map ospf-to-bgp deny 20	

Answer:

Drag the command on the left to the proper order on the right to build a route map to redistribute OSPF prefixes into BGP, so that certain prefixes that are tagged with 65001 are not redistributed.

route-map ospf-to-bgp deny 10	route-map ospf-to-bgp deny 10
route-map ospf-to-bgp permit 10	match tag 65001
match ip tag 65001	route-map ospf-to-bgp permit 20
match tag 65001	
route-map ospf-to-bgp permit 20	
route-map ospf-to-bgp deny 20	

Explanation: route-map ospf-to-bgp deny 10 – first command
match tag 65001 – second command
route-map ospf-to-bgp permit 20 – third command

QUESTION NO: 25

Which MIB supports BGP on Cisco devices?

A. CISCO-BGP4-MIB

- B. IETF-BGP4-MIB
- C. CISCO-ROUTE-MIB
- D. CISCO-IPBGP-MIB

Answer: A

Explanation:

QUESTION NO: 26

Which Cisco IOS Software feature should you enable to monitor IGP neighbor flaps?

- A. IP SLA
- B. adjacency change logging
- C. NetFlow
- D. route dampening
- E. control plane protection

Answer: B

Explanation:

QUESTION NO: 27

Which route-map statement allows further processing of these route-map sections even after the map condition has been met?

- A. goto
- B. continue
- C. jump
- D. follow

Answer: B

Explanation:

QUESTION NO: 28

Refer to the exhibit.

```

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR

Gateway of last resort is not set

 4.0.0.0/24 is subnetted, 1 subnets
D    4.4.4.0 [90/409600] via 161.108.0.4, 00:49:24, Ethernet0/0
 5.0.0.0/24 is subnetted, 1 subnets
C    5.5.5.0 is directly connected, Loopback0
162.108.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    162.108.10.0/24 is directly connected, Serial1/0
C    162.108.4.0/22 is directly connected, Serial2/0
C    161.108.0.0/16 is directly connected, Ethernet0/0
Router#
Router#show ip bgp
BGP table version is 6, local router ID is 5.5.5.5
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop        Metric LocPrf Weight Path
  * 11.1.1.0/24    132.108.10.1      0      100      0 1 1
  * 1131.108.0.0   132.108.10.1      0      100      0 1 1
  *>161.108.0.0    4.4.4.4           0      100      0 1
Router#

```

Data that is destined for 1.1.1.17 is being dropped by this router, though it is in the BGP table. Why are these packets being dropped?

- A. Synchronization is disabled
- B. The next-hop address is not reachable by this router
- C. The AS edge router toward the destination set the next hop to itself
- D. The classful address .0.0.0 has no static route to NULL entry in the IP routing table

Answer: B

Explanation:

QUESTION NO: 29

What is the correct CLI hierarchy when configuring an interface for OSPF in Cisco IOS XR software?

- A. interface, area, router
- B. area, interface, router
- C. router, area, interface
- D. area, router, interface

Answer: C

Explanation:

QUESTION NO: 30

What is the most significant BGP attribute that is used for path selection in Cisco IOS XR Software?

- A. MED
- B. Weight
- C. local preference
- D. router ID

Answer: B

Explanation:

QUESTION NO: 31

How will the syslog report an error condition that is detected in the BGP session?

- A. %BGP-3-BADMASK
- B. %BGP-3-NOTIFICATION:
- C. %BGP-3-INSUFCHUNKS:
- D. %BGP-3-MAXPFXEXCEED:

Answer: B

Explanation:

QUESTION NO: 32

These commands have been issued on all OSPF routers.

```
router ospf 1
```

```
ispf
```

```
exit
```

How do these commands improve the performance of OSPF?

- A. They help OSPF react faster to a local interface failure.
- B. They help OSPF converge faster in reaction to a network event.
- C. They eliminate the OSPF 30-minute LSA refresh and institute independent LSA aging
- D. They save processor and memory resources during network instability by delaying SPF recalculation.

Answer: D

Explanation:

QUESTION NO: 33

To improve BGP performance in your network, you would like to withdraw routes immediately when the next hop of the route becomes unavailable instead of waiting for the scan interval to detect the problem. Which Cisco IOS software command will configure this failure?

- A. bgp inject-map
- B. bgp dampening
- C. neighbor fall-over
- D. import path selection
- E. neighbor update-source

Answer: C

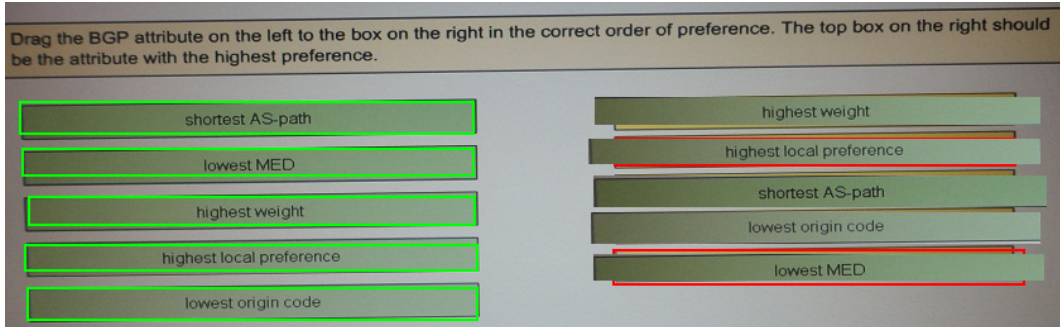
Explanation:

QUESTION NO: 34 DRAG DROP

Drag the BGP attribute on the left to the box on the right in the correct order of preference. The top box on the right should be the attribute with the highest preference.

shortest AS-path	
lowest MED	
highest weight	
highest local preference	
lowest origin code	

Answer:



Explanation: highest weight

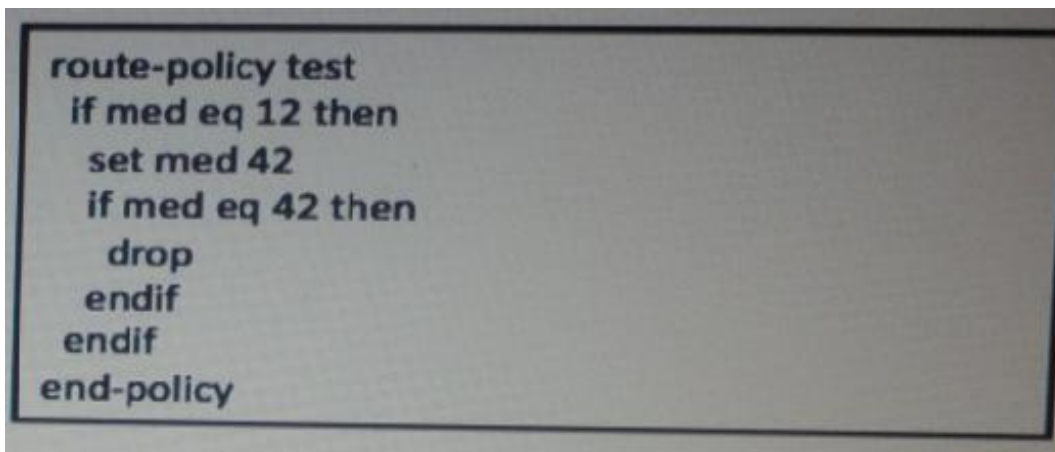
Highest local preference

Shortest AS-path

Lowest origin code

MED

QUESTION NO: 35



Refer to the exhibit. What is the expected output of route-policy “test” if the starting MED value is 12?

- A. set the MED to 42 and then explicit drop the route
- B. set the MED to 42 and then pass the route
- C. drop the route due to implicit drop
- D. set the MED to 42 and then implicit drop the route

Answer: A

Explanation:

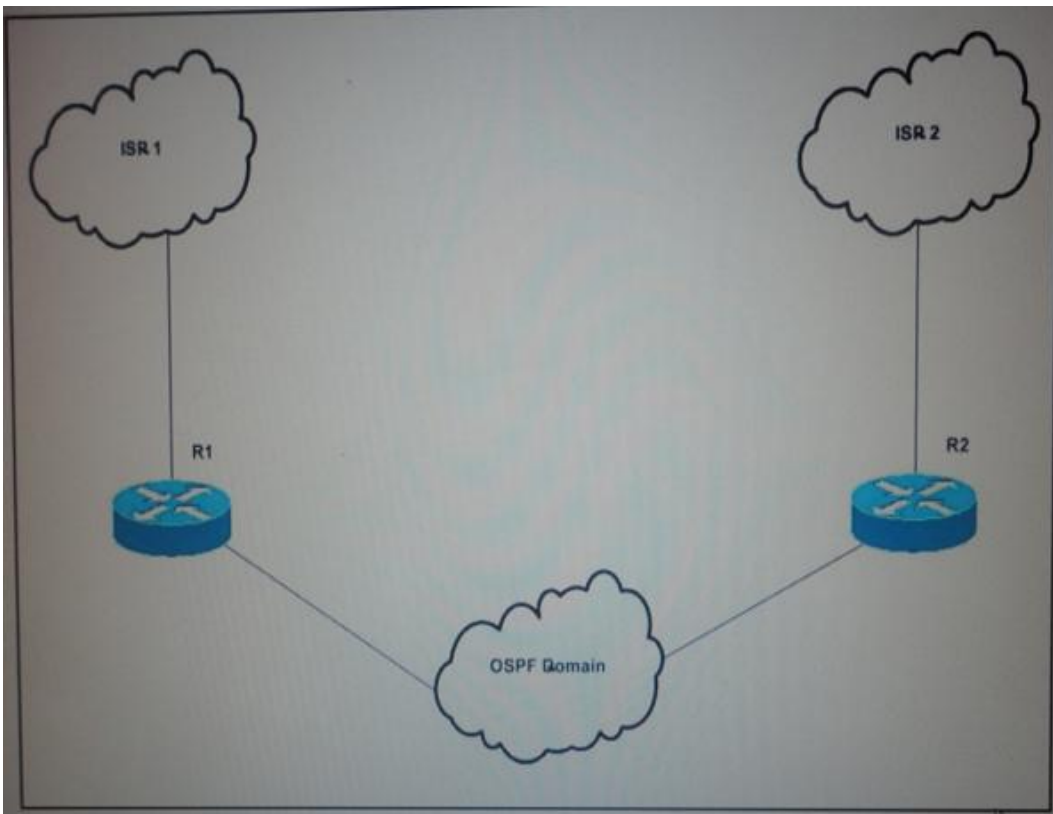
QUESTION NO: 36

Which command displays all interfaces that are configured In OSPF on Cisco IOS XR Software?

- A. show ospf interface
- B. show interface ospf
- C. show ip ospf adj
- D. show ospf database

Answer: A

Explanation:

QUESTION NO: 37

Refer to the exhibit. A client asked you to help solve a problem. This customer has two BGP routers, each connected to its upstream provider. Because of administrative restrictions, these two routers do not have an IBGP session between them, and each router redistributes certain prefixes into OSPF and vice versa. The customer would like to preserve the AS path attribute for prefixes that cross the OSPF domain.

Which route-map set option will let the customer preserve the attribute?

- A. set tag
- B. set as-path
- C. set as-path tag
- D. set extcommunity

Answer: B

Explanation:

QUESTION NO: 38

The service provider has allocated a block of IPv4 addresses for your network 172.16.1.0 with a subnet mask of 25 bits. Your company DHCP server was configured to use this block. According to your company standards, the first available IP address is 172.16.1.130. The NOC has informed you that devices that use this block have partial connectivity. They communicate with other but cannot reach outside networks.

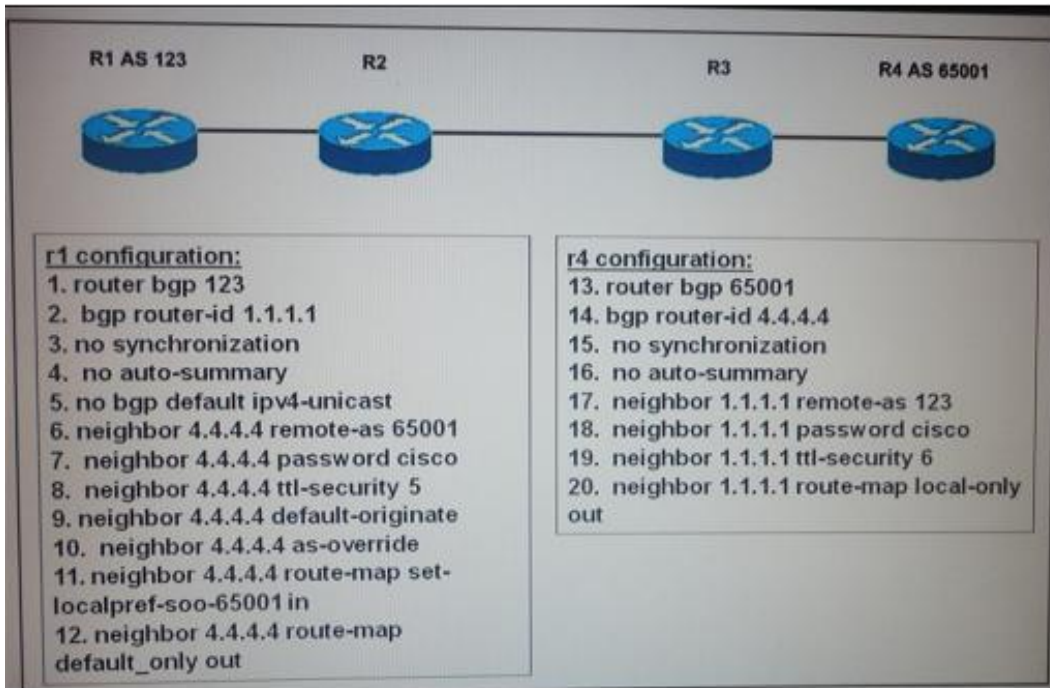
What is the reason for this behavior?

- A. The service provider, by mistake, allocated the RFC 1918 address.
- B. The IP addresses that are distributed by the DHCP server are not on the same subnet as a default gateway
- C. Because everything is configured correctly, the most likely cause is a connectivity issue with a service provider.
- D. Because everything is configured correctly, the most likely cause is a DHCP server issue

Answer: B

Explanation:

QUESTION NO: 39



Refer to the exhibit. Two routers that run Cisco IOS Software cannot establish a BGP session. There is IP connectivity between them. After examining the network topology and router configuration, identify the line number that prevents the establishment of BGP session.

- A. 5
- B. 8
- C. 10
- D. 19

Answer: D

Explanation:

QUESTION NO: 40

Which protocol establishes communications over TCP port 179?

- A. IS-IS
- B. OSPF
- C. EIGRP
- D. BGP
- E. IPsec
- F. PIM

Answer: D

Explanation:

QUESTION NO: 41

```
R2#sh ip bgp
BGP table version is 18, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
Origin codes: i - IGP, e - EGP, ? - incomplete
```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 1.0.0.0	10.1.1.1	0		0 1	i
*> 2.2.2.2/32	0.0.0.0	0		32768	i
*>i3.3.3.3/32	3.3.3.3	0	100	0	i
*>i4.0.0.0	3.3.3.3	0	100	0 4	i
*> 11.0.0.0	10.1.1.1	0		0 1 65111	65112 i
*>i44.0.0.0	3.3.3.3	0	100	0 4 55111	55112 i
*> 111.0.0.0	10.1.1.1	0		0 1	i
*>i144.4.0.0	3.3.3.3	0	100	0 4	i
s> 172.16.4.0/24	10.1.1.1	0		0 1 65201 65203	65204 i
*> 172.16.4.0/22	0.0.0.0			32768	i
s> 172.16.5.0/24	10.1.1.1	0		0 1 65201 65203	65204 i
s> 172.16.6.0/24	10.1.1.1	0		0 1 65201 65203	65204 i
s> 172.16.7.0/24	10.1.1.1	0		0 1 65201 65203	65204 i

Refer to the exhibit. Referring to the output from the show command display, which four statements are correct? (Choose four)

- A. 172.16.4.0/22 is an aggregated route and the summary-only option was used
- B. 2.2.2.2/32 is a local prefix on the R2 router
- C. 1.0.0.0 is a local prefix on the R2 router
- D. 10.1.1.1 is an EBGP peer
- E. 2.2.2.2 is an EBGP peer
- F. 3.3.3.3 is an IBGP peer

Answer: A,B,D,F

Explanation:

QUESTION NO: 42

When using Cisco IOS show commands to examine BGP statistics you noticed that the one of the BGP peers is in the active state. What does that indicate?

- A. The BGP neighborship to the BGP peer is up.
- B. The BGP peer is actively sending III SI' updates.
- C. The BGP session is not established to the BGP peer

D. The 3-way TCP handshake with the BGP peer has successfully completed

Answer: C

Explanation:

QUESTION NO: 43

Which two configuration modes are valid when configuring BGP on Cisco IOS XR Software?
(Choose two)

- A. route-reflector configuration mode
- B. router configuration mode
- C. peer-group configuration mode
- D. neighbor address-family configuration mode

Answer: B,D

Explanation:

QUESTION NO: 44

Which statement about BGP configuration in Cisco IOS XR Software is true?

- A. IPv4 is the default address-family.
- B. VPNv4 is the default address-family.
- C. There is no default address-family.
- D. IPv4 multicast is the default address-family.

Answer: A

Explanation:

QUESTION NO: 45

```

===== r2 =====
*Mar 9 14:04:42.499 UTC: %OSPF-5-ADJCHG: Process 577, Nbr 64.255.255.1
on Serial0/0 from EXSTART to DOWN, Neighbor Down: Too many
retransmissions
*Mar 9 14:05:42.519 UTC: %OSPF-5-ADJCHG: Process 577, Nbr 64.255.255.1
on Serial0/0 from DOWN to DOWN, Neighbor Down: Ignore timer expired
r2#sh ip ospf neighbor
Neighbor ID  Pri  State      Dead Time  Address    Interface
64.255.255.1  0  EXSTART/  -  00:00:39  64.192.12.1  Serial0/0
*Mar 9 14:07:13.139 UTC: OSPF: Retransmitting DBD to 64.255.255.1 on
Serial0/0
*Mar 9 14:07:13.139 UTC: OSPF: Up DBD Retransmit cnt to 18 for
64.255.255.1 on Serial0/0
*Mar 9 14:07:13.139 UTC: OSPF: Send DBD to 64.255.255.1 on Serial0/0 seq
0xE06 opt 0x52 flag 0x7 len 32
*Mar 9 14:07:13.243 UTC: OSPF: Rcv DBD from 64.255.255.1 on Serial0/0
seq 0xE06 opt 0x52 flag 0x2 len 52  mtu 4072 state EXSTART
*Mar 9 14:07:13.243 UTC: OSPF: Nbr 64.255.255.1 has larger interface MTU
===== r1 =====
r1#sh ip ospf neighbor
Neighbor ID  Pri  State      Dead Time  Address    Interface
64.255.255.2  0  EXCHANGE/  -  00:00:38  64.192.12.2  Serial0/0

```

Refer to the exhibit. Two Cisco IOS routers that are configured with OSPF cannot establish adjacency. Examine the information collected from r1 and r2 routers. Why cannot adjacency be established?

- A. network-type mismatch
- B. different MTU
- C. authentication password mismatch
- D. area ID mismatch

Answer: B

Explanation:

QUESTION NO: 46

```

Router#show ip ospf database

  OSPF Router with ID (172.30.1.1) (Process ID 1)

  Router Link States (Area 0)

  <some output deleted for brevity>

  Type-5 AS External Link States

Link ID        ADV Router    Age          Seq#          Checksum Tag
192.168.1.0    192.168.100.1 767         0x80000001   0x007b99 0
192.168.1.0    192.168.100.2 176         0x80000001   0x00f3a2 0
192.168.10.0   192.168.100.1 767         0x80000001   0x001af3 0
192.168.10.0   192.168.100.2 176         0x80000001   0x0090fc 0
192.168.100.1  192.168.100.1 774         0x80000001   0x002e84 0
192.168.100.2  192.168.100.2 183         0x80000001   0x009a96 0

Router#show ip route

  <some output deleted for brevity>

O E1 192.168.10.0/24 [110/174] via 10.255.255.254, 00:04:26, Serial0/0/0
  172.17.0.0/24 is subnetted, 1 subnets
C    172.17.1.0 is directly connected, Ethernet0/0/0
  172.16.0.0/24 is subnetted, 1 subnets
O    172.16.1.0 [110/74] via 10.255.255.254, 00:04:26, Serial0/0/0
  172.31.0.0/32 is subnetted, 1 subnets
O    172.31.1.1 [110/65] via 10.255.255.254, 00:04:26, Serial0/0/0
  172.30.0.0/24 is subnetted, 1 subnets
C    172.30.1.0 is directly connected, Loopback0
  10.0.0.0/30 is subnetted, 1 subnets
C    10.255.255.252 is directly connected, Serial0/0/0
O E1 192.168.1.0/24 [110/174] via 10.255.255.254, 00:04:27, Serial0/0/0
  192.168.100.0/32 is subnetted, 2 subnets
O E2 192.168.100.1 [110/100] via 172.17.1.1, 00:04:36, Ethernet0/0/0
O E1 192.168.100.2 [110/174] via 10.255.255.254, 00:04:36, Serial0/0/0

  <some output deleted for brevity>

```

Refer to the exhibit. The OSPF database shows that there are two OSPFs 92.168.1.0. What caused the router to install the one particular path in the routing table instead of the other?

- A. The administrative distance is lower.
- B. The path not being used has aged out.
- C. E1 routes take precedence over E2 routes.
- D. The chosen path has the smallest OSPF metric
- E. Intra-area routes are preferred over interarea routes

Answer: A

Explanation:

QUESTION NO: 47

```

route-policy inner($med)
  if community matches-every (1583:241, 1583:1241, 1583:4241)
    set med $med
  else set med 75
end-policy
|
route-policy test
  apply inner (50)
end-policy

```

Refer to the exhibit. What will be the multi-exit-discriminator value for a route with community values of 1583.241 1583:4241 if the Cisco IOS XR Software Route Policy Language policy "test" is applied?

- A. 50
- B. 75
- C. 125
- D. will not change

Answer: B

Explanation:

QUESTION NO: 48

```

PE1#show run | begin router ospf 1
router ospf 1
 log-adjacency-changes
 redistribute static subnets
 network 10.0.0.0 0.255.255.255 area 0
 network 192.168.0.0 0.0.255.255 area 0
!
<output omitted>

PE1#show interfaces GigabitEthernet0/0
GigabitEthernet0/0 is up, line protocol is up
 Hardware is MV64460 Internal MAC, address is 0026.cbf5.a31b
 Description: Link to P1
 Internet address is 192.168.143.44/24
 MTU 4500 bytes, BW 1000000 Kbit, DLY 10 usec,
 reliability 255/255, txload 1/255, rxload 1/255
 Encapsulation ARPA, loopback not set
 Keepalive set (10 sec)
 Full Duplex, 100Mbps, media type is RJ45
 output flow-control is XON, input flow-control is unsupported
 ARP type: ARPA, ARP Timeout 04:00:00
 Last input 00:00:05, output 00:00:00, output hang never
 Last clearing of "show interface" counters never
 Input queue: 0/75/0/0 (size/max/drops/flushes): Total output drops: 0
 Queueing strategy: fifo
 Output queue: 0/40 (size/max)
 5 minute input rate 0 bits/sec, 0 packets/sec
 5 minute output rate 0 bits/sec, 0 packets/sec
 209810 packets input, 24838502 bytes, 0 no buffer
 Received 191672 broadcasts (0 IP multicasts)
 0 runs, 0 giants, 0 throttles
 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
 0 watchdog, 234170 multicast, 0 pause input
 245364 packets output, 25638651 bytes, 0 underruns
 2 output errors, 0 collisions, 1 interface resets
 0 bables, 0 late collision, 0 deferred
 2 lost carrier, 0 no carrier, 0 pause output

RP/0/0/CPU0: PE4#show run | begin router ospf 1
router ospf 1
 address-family ipv4 unicast
 area 0
 dead-interval 20
 hello-interval 5
 interface Loopback0
 passive enable
!
 interface GigabitEthernet0/4/0/2
 cost 1000
!
 interface GigabitEthernet0/4/0/3
 cost 1000
!
<output omitted>

RP/0/0/CPU0: PE4#show interfaces GigabitEthernet0/4/0/2
GigabitEthernet0/4/0/2 is up, line protocol is up
 Interface state transitions: 16
 Hardware is GigabitEthernet/IEEE 802.3 interface(s), address is
 0018.18c7.4792
 (bia 0018.18c7.4792)
 Internet address is 192.168.143.45/24
 MTU 4514 bytes, BW 1000000 Kbit
 reliability 255/255, txload 0/255, rxload 0/255
 Encapsulation ARPA,
 Full-duplex, 1000Mb/s, SFPD, link type is force-up
 output flow control is on, input flow control is off
 loopback not set
 ARP type ARPA, ARP timeout 04:00:00
 Last clearing of "show interface" counters never
 5 minute input rate 0 bits/sec, 0 packets/sec
 5 minute output rate 0 bits/sec, 0 packets/sec
 0 packets input, 0 bytes, 0 total input drops
 0 drops for unrecognized upper-level protocol
 Received 0 broadcast packets, 0 multicast packets
 0 runs, 0 giants, 0 throttles, 0 parity
 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
 60 packets output, 13560 bytes, 0 total output drops
 Output 0 broadcast packets, 60 multicast packets
 0 output errors, 0 underruns, 0 applique, 0 resets
 0 output buffer failures, 0 output buffers swapped out
 15 carrier transitions

```

Refer to the exhibit The Cisco IOS Software platform PE1 has interface GigabitEthernet port 0/0 connected to the Cisco IOS XR software platform PE4 interface GigabitEthernet 0/4/0/2. OSPF cannot establish an adjacency. What is a possible solution to the problem?

- A. Explicitly configure the Cisco IOS XR Software interface to be active OSPF interface
- B. Configure OSPF on each platform to ignore the interface MTU
- C. Adjust the Cisco IOS XR Software hello and dead times to match Cisco IOS software
- D. Change the interface configurations on both platforms to point network type

Answer: C

Explanation:

QUESTION NO: 49

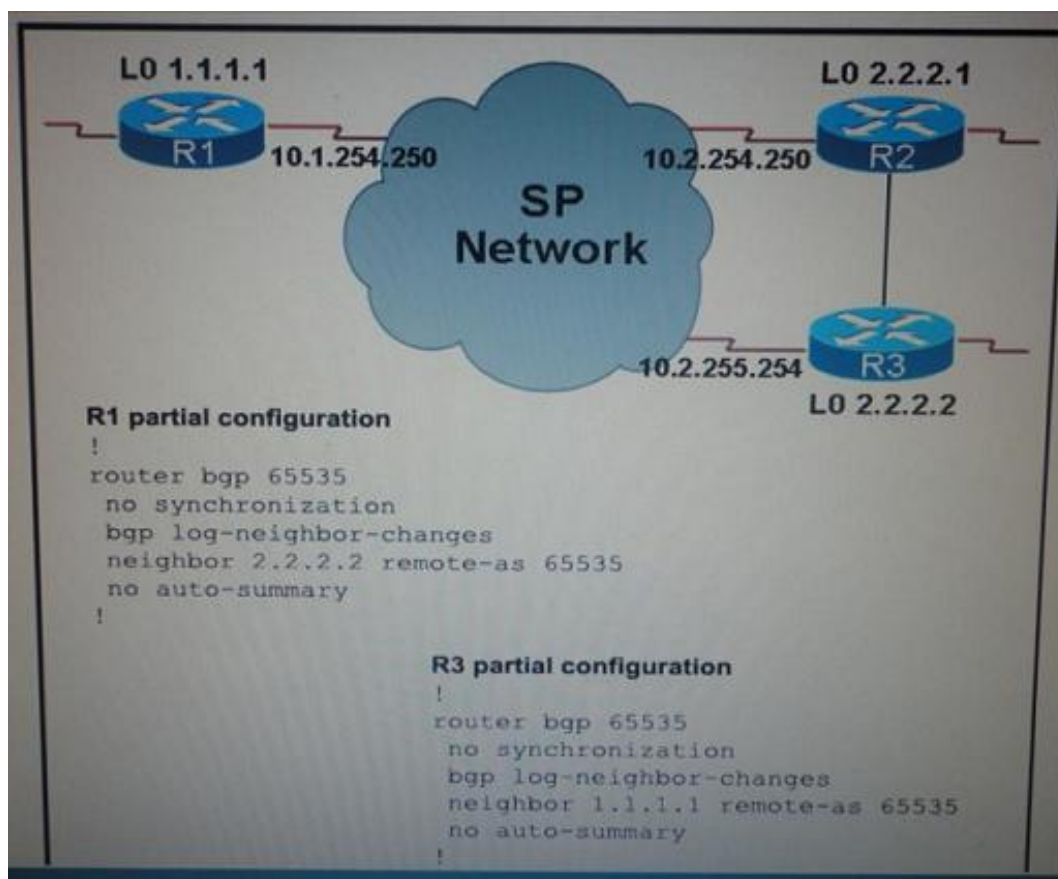
What is BGP CEF purge time on a CRS-1?

- A. the time required to remove BGP routes from the FIB after clearance notification
- B. the time required to remove BGP routes from the RIB after clearance notification
- C. the time required to declare a BGP neighbor as down
- D. the time required to remove routes from the BGP routing table

Answer: D

Explanation:

QUESTION NO: 50



Refer to the exhibit. R1 and R3 are trying to peer with each other, but the session will not establish. What can be done to allow BGP to establish the session?

- A. Both routers should set the BGP next hop as self.
- B. Each router must explicitly activate the session before it can be established.
- C. On both routers, specify that the source of the session is the loopback interface.
- D. Increase the number of hops that are allowed when building the BGP session between the routers.

Answer: C,D

Explanation:

QUESTION NO: 51

What does the SNMP object "ospfNbrTable" provide?

- A. a table of nonvirtual neighbor information
- B. a list of neighbors for the OSPF interface
- C. area border list of neighbors
- D. a list of OSPF-to-NetBIOS exchanges

Answer: A

Explanation:

QUESTION NO: 52

Which Cisco IOS Software command lengthens the BGP update time reduce the number of updates that are sent when flapping BGP links are causing network instability?

- A. timers bgp
- B. bgp dampening
- C. neighbor timers
- D. bgp aggregate-timer
- E. neighbor advertisement-interval

Answer: B

Explanation:

QUESTION NO: 53

Which Cisco IOS XR Software Route Policy Language policy will set the weight to 50 for routes that match the prefix-set "test"?

- A.** route-policy new
if ip-address in test then
set weight 50
endif
end-policy
- B.** route-policy new
match ip address prefix-list test
set weight 50
end-policy
- C.** route-policy new
if destination in test then
set weight 50
endif
end-policy
- D.** route-policy new
match prefix-list test
set weight 50
end-policy

Answer: C

Explanation:

QUESTION NO: 54

To influence the inbound traffic flow into an AS, which BGP attribute would signal an external AS to use the preferred path unto the AS?

- A.** local preference
- B.** MED
- C.** Weight
- D.** next-hop
- E.** origin

Answer: B

Explanation:

QUESTION NO: 55

Output from the show ip bgp command shows the state of the router IBGP sessions to be active. Which of the following statements may be true?

- A. No neighbors have been configured
- B. Connections to the neighbors are up and operational
- C. The router cannot find its neighbors in its IGP routing table
- D. Prefixes to advertise have not been specified with the network command.

Answer: C

Explanation:

QUESTION NO: 56

In an effort to decrease BGP convergence time, the BGP scan timer has been lowered. What is possible side effect of this action?

- A. There are slow or no responses to the console or Telnet sessions
- B. Maximum packet-per-second throughput will increase.
- C. Batch updates will be held for a shorter time
- D. The BGP table can become out of sync with the rest of the network

Answer: A

Explanation:

QUESTION NO: 57

Refer to the exhibit.

```
route-policy LONGPREF
  if destination in(0.0.0.0/0 ge 28) then
    apply NESTED(222, 333)
  else
    apply NESTED(777, 888)
  endif
end-policy
route-policy NESTED($var1, $var2)
  if destination in(0.0.0.0/0 le 32) and (community matches-any (999:999) or community is-empty) then
    set community($var1:$var2)
  endif
end-policy
```

You are advertising prefix 71.44.1.180/31 with an unset community to an eBGP neighbor. After applying route policy LONGPREF in the outbound direction to the eBGP. What will the community be for the advertised prefix?

- A. will be 222:333
- B. will be 777:888
- C. will be 1:2
- D. will be 999:999
- E. will be empty
- F. prefix will not be passed

Answer: A

Explanation:

QUESTION NO: 58

Which way can you distribute BGP processes to achieve better scalability on CRS-1 routers that run Cisco IOS XR software?

- A. between the active RP and the standby RP
- B. between the active RP and the MSBs
- C. between the active RP and the DRPs
- D. between the active RP and the PRPs

Answer: C

Explanation:

QUESTION NO: 59

What will be the result of the aggregate-address 192.168.0.0 255.255.0.0 BGP command?

- A. The router that is doing the aggregation will advertise the 192.168.0.0/16 summary route as long as there is at least one specific route within the range in the BGP table
- B. The more specific routes will be suppressed and only the 192.168.0.0/16 summary route will be advertised
- C. The BGP table size on the router that is performing the route aggregation will be reduced
- D. The flap of the more specific route will not be visible to the rest of the network

Answer: A

Explanation:

QUESTION NO: 60

What type of LSA will be used by an OSPF ABR to advertise internal routes from a standard area to a

backbone area?

- A. type 1 router LSAs if -ill prefixes are advertised, or as type 3 summary LSAs if the prefixes are summarized
- B. type 2 network LSAs
- C. type 3summary LSAs, whether or not the prefixes are summarized
- D. type 5 external LSAs

Answer: C

Explanation:

QUESTION NO: 61

Without using route reflectors or confederations, an AS with 20 routers will require how many IBGP peerings?

- A. 20
- B. 45
- C. 190
- D. 200
- E. 400

Answer: C

Explanation:

QUESTION NO: 62

Which three statements correctly describe the BGP route-flap-dampening feature?

(Choose three)

- A. Each BGP route flap is penalized by adding 1000 penalty points to the penalty
- B. If a cumulative penalty exceeds the suppress limit, the route is dampened.
- C. The penalty is gradually decreased after the route stops flapping. When the penalty reaches one half of the reuse limit, the route is removed from the BGP table.
- D. When a penalty drops below the reuse limit or when the route has been dampened for more than the maximum suppress time, the route becomes valid again.
- E. The penalty reduction rate is determined by the reuse limit.

Answer: A,B,D

Explanation:

QUESTION NO: 63

Which statement is correct regarding the BGP Fast Peering Session Deactivation feature?

- A.** improves the BGP convergence time by faster detection of router flaps
- B.** lowers the CPU usage by reducing the BGP scan time
- C.** allows BGP to detect an adjacency change and deactivate the terminated session in between standard CGP scanning intervals
- D.** allows a Cisco NSF-aware router to assist Cisco NSF-capable and Cisco NSF-aware neighbor to continue forwarding packets during a switchover operation

Answer: C

Explanation:

QUESTION NO: 64

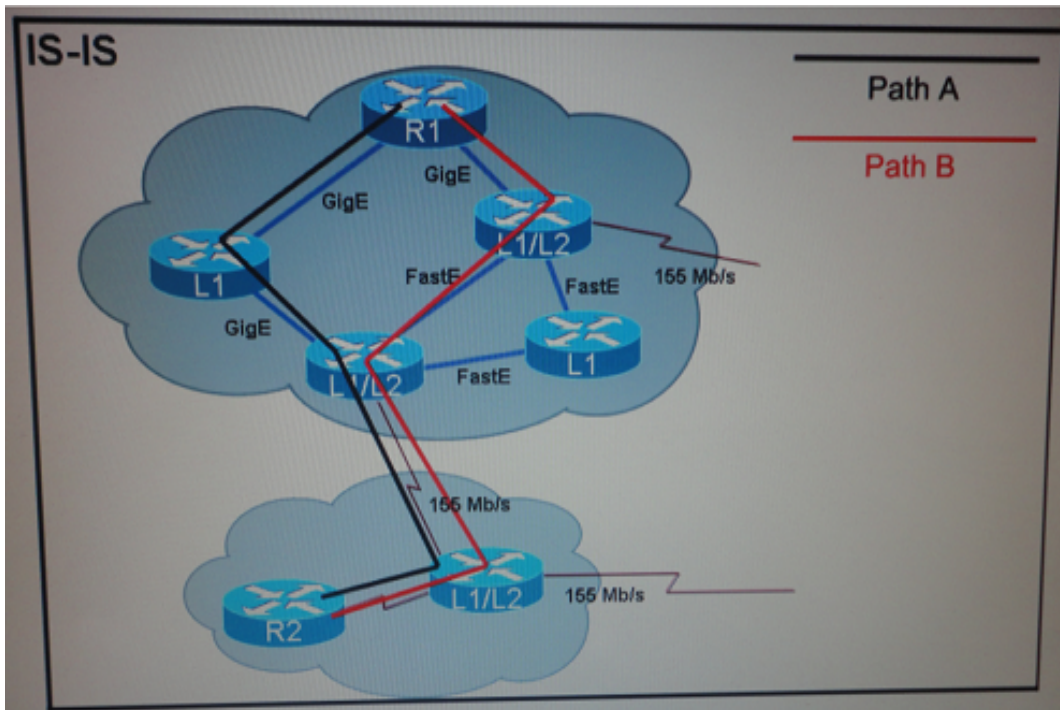
What is the difference between community sets and extended community sets in Cisco IOS XR Software?

- A.** Extended community sets match community values that are larger than 65535.
- B.** Extended community sets match non-IP traffic.
- C.** Extended community sets are used when all regular community sets are in use.
- D.** Extended community sets are used to match extended community values.

Answer: D

Explanation:

QUESTION NO: 65



Refer to the exhibit. Which ISIS feature needs to be configured to solve the incorrect load balancing of data traffic?

- A. Tag the desired route by using the 'isis' tag interface command on R1 Gigabit Ethernet interface.
- B. Set the desired path to a higher priority by using the ip route priority high router level command.
- C. Adjust the interface metrics to reflect the link bandwidths by using the 'isis metric interface' command
- D. Set the link between R1 and the L1/L2 router to a level 1-only link by using the isis circuit-type interface command

Answer: D

Explanation:

QUESTION NO: 66

Which is true when troubleshooting Cisco IOS XR Software by using the CLI?

- A. Debug commands are less CPU-intensive than trace commands.
- B. Trace uses a buffer that is shared by all processes
- C. Trace command logs will survive a system reboot, but debug will not
- D. Both trace and debug features need to be manually enabled when troubleshooting

Answer: D

Explanation:

QUESTION NO: 67

Which commands configure a Cisco IOS XR router to match an AS set value that starts with 400?

- A.** as-set test
match as A400_
end-set
- B.** as -p ath -set test
ios-regex '^400_'
end-set
- C.** as-set test
ios-regex ^400_
end-set
- D.** as-path-set test
as ^400_
end-set

Answer: B

Explanation:

QUESTION NO: 68

Your NMS server reports the message

%SNMP-4-NOTRAPIP: SNMP trap source Gigabit Ethernet 1/0/1 has no ip address

What does this error message indicate?

- A.** The interface to be used as the source address has no address associated with it
- B.** The SNMP server does not have a logging source interface defined.
- C.** The SNMP-server host command has no specified trap destination.
- D.** The interface of the trap destination is missing an address.

Answer: A

Explanation:

QUESTION NO: 69

Two Cisco IOS routers with OSPF configured are unable to establish adjacency. Which debug command will help you to identify the root cause of the problem?

- A. debug ip ospf timers
- B. debug ip ospf neighbors
- C. debug ip ospf network-type
- D. debug ip ospf adjacency

Answer: D

Explanation:

QUESTION NO: 70

You want to filter certain prefixes that are advertised by the BGP neighbor of the neighboring autonomous system. Which command will filter prefixes that are advertised from the BGP neighbor 2.2.2.2? For filtering, use the already configured route map as123.

- A. neighbor 2.2.2.2 route-map as123
- B. neighbor 2.2.2.2 route-map as123 out
- C. neighbor 2.2.2.2 route-map as123 in
- D. neighbor 2.2.2.2 distribute-list as123
- E. neighbor 2.2.2.2 distribute-list as123 out
- F. neighbor 2.2.2.2 distribute-list as123 in

Answer: C

Explanation: